# JUNE CAGAN

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## EDUCATION

**University of Maryland, College Park** *August 2023 - May 2027*
B.S. in Computer Science
Minor in Linguistics

Overall GPA: 3.62

Coursework: Data Structures, Algorithms, Object-Oriented Programming I/II, Linear Algebra, Computer Systems, Compilers, Databases, Web Application Development

## TECHNOLOGIES

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| **Programming Languages:** | Python, C, C++, Rust, OCaml, Java, x86 Assembly, Go, Kotlin, SQL |
| **Web Development:** | JavaScript, HTML, CSS, Node.js, Express, React, MongoDB, REST APIs |
| **Tools & Technologies:** | Git, Claude Code, Docker, CMake, JUnit, npm, Codex |
| **Operating Systems:** | Have deployed and managed Ubuntu and Windows 10/11 |

## EXPERIENCE

**Undergraduate Teaching Assistant** January 2025 - Present
*University of Maryland Department of Computer Science* *College Park, MD*

- Teaching assistant for CMSC330, a class that teaches different paradigms in programming, such as functional programming in OCaml and memory safety through Rust.
- Organized and developed programming exercises for over 850 students, with projects testing students on knowledge of higher-order functions, finite automata, parsing, interpreters, type-checking and garbage collection.
- Led discussion sections weekly, reinforcing core concepts and completing programming exercises with over 30 students.

**Distributed Systems Engineering Intern** June 2026 - August 2026
*Tradeweb* *New York, NY*

- Optimized the algorithm powering the Sweep-AME engine, increasing total notional by up to 1.5% in internal testing and simulation.
- Used CMake, CTest, and GDB to build, test, and debug C++ implementations.
- Expanded unit test coverage to stress-test optimized implementations across diverse inputs and edge cases.

**REU-CAAR Fellow** June 2025 - August 2025
*University of Maryland Department of Computer Science* *College Park, MD*

- Conducted research on geometric algorithms for machine learning and quantum computing applications.
- Formulated new algorithms for the creation of geometric data structures, such as Voronoi diagrams and Delaunay triangulations, in the Hilbert metric space.
- Worked closely with one researcher to develop software that visualizes algorithms in multiple non-euclidean metrics.

## PROJECTS

**Voronoise** | ***C++, JUCE, CMake***

- Developed a VST3/standalone synthesizer plugin in which balls bounce off user-placed geometric structures, and each collision plays a note.
- Moved the physics engine to the real-time audio thread with lock-free queues and no memory allocation, cutting note timing jitter from up to 24 ms to sample-accurate.
- Wrote the specs, reviewed and debugged generated C++, and used unit and audio-timing tests to verify timing and audio output.

**Hilbert Metric Visualizer** | ***JavaScript, Node.js, Express***

- Made for REU-CAAR 2025 to explore non-euclidean geometries and algorithms.
- Worked on data structures, object and algorithm design, and user interaction.
- Used Node.js and Express for backend development, utilizing the Canvas API for graphics.